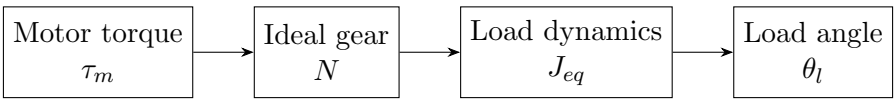
One-DOF Teleoperation: Horizontal Baseline

Model and Control Design Notebook

1 System Architecture

Topology: Single-link revolute joint; ideal gear; horizontal XY plane.



Symbol	Meaning	Value / unit	Status
L	Link length	0.150 m	Design target
N	ω_m/ω_l	69255/208	Catalog; nominal 333:1
$\tau_{g,c}$	Gear continuous rating	1.8 N m	Catalog output limit
η	Gear efficiency	1	Ideal assumption

$$\theta_m = N\theta_l, \qquad \omega_m = N\omega_l, \qquad \tau_{l,m} = N\tau_m$$

$$\mathbf{g} = -g\mathbf{e}_z, \qquad \tau_{g,z} = (\mathbf{r}_c \times m\mathbf{g}) \cdot \mathbf{e}_z = 0$$

Scope: No friction, backlash, compliance or contact. Vertical legacy model retained.

Selected Gearhead: maxon GP 22 C

Symbol	Meaning	Value / unit	Status
η_{max}	Maximum efficiency	0.49	Catalog; not simulated
J_g	Input-side gear inertia	4×10^{-8} kg m^2	Catalog; included

Source: maxon, GP 22 C catalog, March 2025, pp. 421–422. Exact ratio $69255/208 \simeq 332.9567$. EC-max 16 appears in the modular combinations.

$$\tau_{out,steady} \leq \min\{1.8,\ \eta N \tau_{m,continuous}\} \quad [\mathrm{N\,m}]$$

Qualification: Steady forward-drive estimate only. Motor thermal limits, operating efficiency and assembly variant remain to be verified. The ideal baseline uses $\eta = 1$; catalog efficiency is not a friction model.

$$J_{m\to l} = N^2 J_m \simeq 9.423 \times 10^{-3}\ \mathrm{kg\,m^2}$$

Haptic implication: Reflected rotor inertia increases. Backdrivability requires separate assessment.

2 Mechanical Model

Symbol	Meaning	Value / unit	Status
J_l	Link inertia	2.8024×10^{-4} kg m^2	Calculated; assumed geometry
J_m	Rotor inertia	0.85×10^{-7} kg m^2	Legacy; unverified
τ_m	Motor torque	N m	Prescribed input
τ_{ext}	External load torque	N m	Positive with θ_l

$$T = \frac{1}{2} J_l \dot{\theta}_l^2 + \frac{1}{2} J_m (N\dot{\theta}_l)^2, \qquad J_{eq} = J_l + N^2 J_m$$

$$\boxed{J_{eq}\ddot{\theta}_l = N\tau_m + \tau_{ext}}$$

$$\boxed{\frac{\Theta_l(s)}{T_m(s)} = \frac{N}{J_{eq}s^2}}, \qquad \frac{\Theta_l(s)}{T_{ext}(s)} = \frac{1}{J_{eq}s^2}$$

Boundary: Motor torque is ideal. Electrical dynamics and actuator limits are not yet represented.

150 mm design geometry $\not\Rightarrow$ verified legacy inertia

Rectangular Link Geometry

Symbol	Meaning	Value / unit	Status
L	Length	0.150 m	Agreed target
w, h	Width, thickness	$0.020, 0.010$ m	Assumed
ρ	Density	1240 kg m^{-3}	Assumed solid PLA
m	Mass	0.0372 kg	Calculated
l_c	COM distance	0.075 m	Calculated

$$m = \rho L w h, \qquad I_{zz,c} = \frac{m}{12}(L^2 + w^2), \qquad I_{zz,p} = I_{zz,c} + m(L/2)^2$$

Axis: Length along X; joint about Z. Real print mass and inertia remain unverified.

3 Control Design Interface

Symbol	Meaning	Value / unit	Status
ω_i	Current-loop bandwidth	TBD / rad s^{-1}	Pending limits
ω_v	Speed-loop bandwidth	TBD / rad s^{-1}	Pending design
K_t	Torque constant	TBD / N m A^{-1}	Verify motor variant
R, L_a	Resistance, inductance	TBD / Ω, H	Verify motor variant

Planned topology: Current PI; load-speed PI; load-position P.

$$L_a\dot{i} + Ri + K_e N\dot{\theta}_l = v, \qquad \tau_m = K_t i$$

$$P_i(s) = \frac{1}{L_a s + R}, \qquad C_i(s) = \frac{\omega_i(L_a s + R)}{s}$$

$$P_v(s) = \frac{N}{J_{eq} s} \quad \text{(motor-torque input, load-speed output)}$$

Status: Equations are design interfaces, not implemented controllers. No numerical gains are approved. Finite current-loop bandwidth and saturation require verification.

$$\theta_m - \theta_s \to 0, \qquad \tau_h + \tau_e \to 0$$

Bilateral stage: Differential/common modes and DOB follow single-axis validation.

4 Archived Reduced-Model Verification

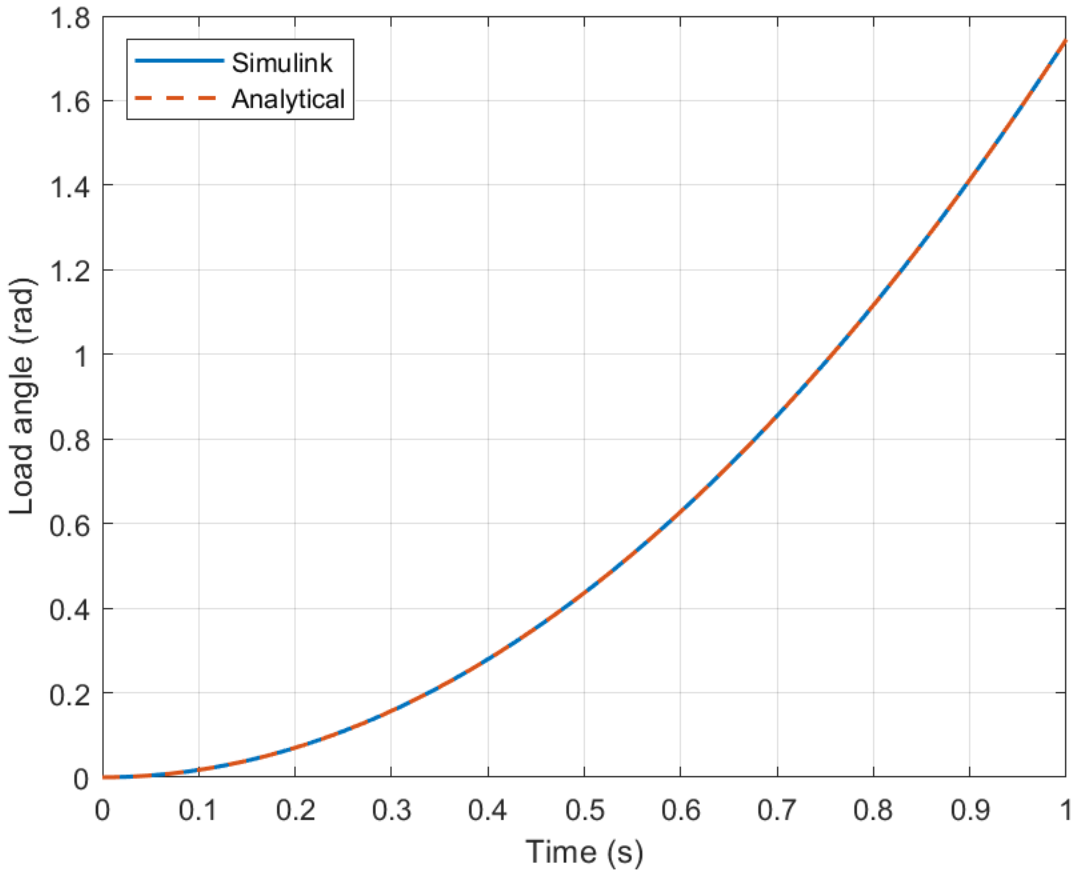
Status: Historical ideal-inertia check only. The generated reduced model was removed; this is not validation of the retained motor–link model.

Symbol	Meaning	Value / unit	Status
τ_0	Motor torque step	10^{-4} N m	Test input
t_f	Duration	1 s	Test setting
N	Ratio	69255/208	Selected; idealized
ϵ_θ	Maximum angle error	$< 10^{-6}$ rad	Acceptance criterion

$$\theta_l(0) = 0, \quad \dot{\theta}_l(0) = 0, \quad \tau_{ext} = 0$$

$$\boxed{\theta_l(t) = \frac{N\tau_0}{2J_{eq}} t^2}$$

$$e_{max} = \max_k |\theta_{sim}(t_k) - \theta_l(t_k)|$$



Next: Confirm motor winding and assembly product codes; verify inertia and torque limits.

5 Cascade Bandwidth Design

Symbol	Meaning	Value / unit	Status
ω_i	Current target	31415.93 rad/s	Retained nominal target
ω_v	Load-speed target	3141.59 rad/s	Retained nominal target
K_x	Position gain	6.2832 s^{-1}	User setting
B_l	Load-side damping	$N^2B_m + B_g$	Linear viscous model

$$P_i = \frac{1}{L_a s + R}, \quad C_i = \frac{\omega_i(L_a s + R)}{s}, \quad T_i = \frac{\omega_i}{s + \omega_i}$$

$$P_v = \frac{N}{Js + B_l}, \quad C_v = \frac{\omega_v(Js + B_l)}{Ns}$$

$$K_{p,v} = \frac{J\omega_v}{N}, \quad K_{i,v} = \frac{B_l\omega_v}{N}, \quad \frac{K_{i,v}}{K_{p,v}} = \frac{B_l}{J}$$

$$L_v^{ideal} = \frac{\omega_v}{s}, \quad L_v = C_v T_i P_v, \quad T_v = \frac{L_v}{1 + L_v}, \quad L_x = \frac{K_x T_v}{s}, \quad T_x = \frac{L_x}{1 + L_x}$$

Bode: Open-loop 0 dB crossover versus closed-loop −3 dB bandwidth. Position P control does not cancel a pole. **Limits:** Nominal LTI model; gravity, saturation, sampling and residual back-EMF coupling excluded. Current and speed targets require actuator feasibility assessment.

